

```
In [1]: import pandas as pd
import os
from sqlalchemy import create_engine
```

```
In [2]: from sqlalchemy import create_engine

engine = create_engine(
    "mysql+pymysql://root:Tvs%405755@localhost:3306/inventory"
)
```

```
In [4]: for file in os.listdir('data'):
    if '.csv' in file:
        df=pd.read_csv('data/'+file)
        print(df.shape)
        ingest_db(df,file[:-4],engine)
```

```
(206529, 9)
(224489, 9)
(2372474, 16)
(12261, 9)
(6239247, 14)
(5543, 10)
```

```
In [3]: ##ingest_db() is a user-defined function that stores a Pandas DataFrame into a SQL
def ingest_db(df, table_name, engine):
    df.to_sql(table_name, con=engine, if_exists='replace', index=False)
```

```
In [5]: tables = pd.read_sql("SHOW TABLES;", engine)
print(tables)
```

```
Tables_in_inventory
0    begin_inventory
1    end_inventory
2    purchase_prices
3    purchases
4    sales
5    vendor_invoice
```

```
In [9]: import pandas as pd

pd.read_sql("SELECT COUNT(*) FROM purchases", engine)
```

```
Out[9]:    COUNT(*)
0      2372474
```

```
In [11]: tables = pd.read_sql("SHOW TABLES;", engine)
table_col = tables.columns[0]

for table in tables[table_col]:
    count = pd.read_sql(
        f"SELECT COUNT(*) AS count FROM {table}",
        engine
```

```

    )['count'][0]

    print(table, ":", count)

```

```

begin_inventory : 206529
end_inventory   : 224489
purchase_prices : 12261
purchases       : 2372474
sales           : 6239247
vendor_invoice  : 5543

```

```

In [12]: tables = pd.read_sql("SHOW TABLES;", engine)
         print(tables)

```

```

Tables_in_inventory
0    begin_inventory
1    end_inventory
2    purchase_prices
3    purchases
4    sales
5    vendor_invoice

```

```

In [13]: for table in tables[tables.columns[0]]:
         print(table)

```

```

begin_inventory
end_inventory
purchase_prices
purchases
sales
vendor_invoice

```

```

In [14]: tables = pd.read_sql("SHOW TABLES;", engine)
         table_col = tables.columns[0]

         for table in tables[table_col]:
             count = pd.read_sql(
                 f"SELECT COUNT(*) AS count FROM {table}",
                 engine
             ).iloc[0, 0]

             print(f"{table}: {count}")

```

```

begin_inventory: 206529
end_inventory: 224489
purchase_prices: 12261
purchases: 2372474
sales: 6239247
vendor_invoice: 5543

```

```

In [16]: import pandas as pd
         # Get all table names
         tables = pd.read_sql("SHOW TABLES;", engine)
         # Get the table name column
         table_col = tables.columns[0]
         # Loop through each table
         for table in tables[table_col]:
             print("-"*50, table, "-"*50)

```

```
# Count records
count = pd.read_sql(
    f"SELECT COUNT(*) AS count FROM {table}",
    engine
)['count'].iloc[0]
print("Count of records:", count)
# Display first 5 rows
display(pd.read_sql(
    f"SELECT * FROM {table} LIMIT 5",
    engine
))
```

```
----- begin_inventory -----
```

Count of records: 206529

	InventoryId	Store	City	Brand	Description	Size	onHand	Price	status
59	1_HARDERSFIELD_58	1	HARDERSFIELD	58	Gekkeikan Black & Gold Sake	750mL	8	12.99	2024-08-26 14:00
60	1_HARDERSFIELD_60	1	HARDERSFIELD	60	Canadian Club 1858 VAP	750mL	7	10.99	2024-08-26 14:00
61	1_HARDERSFIELD_62	1	HARDERSFIELD	62	Herradura Silver Tequila	750mL	6	36.99	2024-08-26 14:00
62	1_HARDERSFIELD_63	1	HARDERSFIELD	63	Herradura Reposado Tequila	750mL	3	38.99	2024-08-26 14:00
63	1_HARDERSFIELD_72	1	HARDERSFIELD	72	No. 3 London Dry Gin	750mL	6	34.99	2024-08-26 14:00



```
----- end_inventory -----
```

Count of records: 224489

	InventoryId	Store	City	Brand	Description	Size	onHand	Price	en
0	1_HARDERSFIELD_58	1	HARDERSFIELD	58	Gekkeikan Black & Gold Sake	750mL	11	12.99	
1	1_HARDERSFIELD_62	1	HARDERSFIELD	62	Herradura Silver Tequila	750mL	7	36.99	
2	1_HARDERSFIELD_63	1	HARDERSFIELD	63	Herradura Reposado Tequila	750mL	7	38.99	
3	1_HARDERSFIELD_72	1	HARDERSFIELD	72	No. 3 London Dry Gin	750mL	4	34.99	
4	1_HARDERSFIELD_75	1	HARDERSFIELD	75	Three Olives Tomato Vodka	750mL	7	14.99	



----- purchase_prices -----

Count of records: 12261

	Brand	Description	Price	Size	Volume	Classification	PurchasePrice	VendorNumber
0	58	Gekkeikan Black & Gold Sake	12.99	750mL	750	1	9.28	8320
1	62	Herradura Silver Tequila	36.99	750mL	750	1	28.67	1128
2	63	Herradura Reposado Tequila	38.99	750mL	750	1	30.46	1128
3	72	No. 3 London Dry Gin	34.99	750mL	750	1	26.11	9165
4	75	Three Olives Tomato Vodka	14.99	750mL	750	1	10.94	7245



----- purchases -----

Count of records: 2372474

	InventoryId	Store	Brand	Description	Size	VendorNumber	VendorName
0	69_MOUNTMEND_8412	69	8412	Tequila Ocho Plata Fresno	750mL	105	ALTAMAR BRANDS LLC
1	30_CULCHETH_5255	30	5255	TGI Fridays Ultimte Mudslide	1.75L	4466	AMERICAN VINTAGE BEVERAGE
2	34_PITMERDEN_5215	34	5215	TGI Fridays Long Island Iced	1.75L	4466	AMERICAN VINTAGE BEVERAGE
3	1_HARDERSFIELD_5255	1	5255	TGI Fridays Ultimte Mudslide	1.75L	4466	AMERICAN VINTAGE BEVERAGE
4	76_DONCASTER_2034	76	2034	Glendalough Double Barrel	750mL	388	ATLANTIC IMPORTING COMPANY



----- sales -----

Count of records: 6239247

	InventoryId	Store	Brand	Description	Size	SalesQuantity	SalesDollars	Sale
0	1_HARDERSFIELD_1004	1	1004	Jim Beam w/2 Rocks Glasses	750mL	1	16.49	
1	1_HARDERSFIELD_1004	1	1004	Jim Beam w/2 Rocks Glasses	750mL	2	32.98	
2	1_HARDERSFIELD_1004	1	1004	Jim Beam w/2 Rocks Glasses	750mL	1	16.49	
3	1_HARDERSFIELD_1004	1	1004	Jim Beam w/2 Rocks Glasses	750mL	1	14.49	
4	1_HARDERSFIELD_1005	1	1005	Maker's Mark Combo Pack	375mL 2 Pk	2	69.98	



----- vendor_invoice -----

Count of records: 5543

	VendorNumber	VendorName	InvoiceDate	PONumber	PODate	PayDate	Quantity	D
0	105	ALTAMAR BRANDS LLC	2024-01-04	8124	2023-12-21	2024-02-16	6	2
1	4466	AMERICAN VINTAGE BEVERAGE	2024-01-07	8137	2023-12-22	2024-02-21	15	1
2	388	ATLANTIC IMPORTING COMPANY	2024-01-09	8169	2023-12-24	2024-02-16	5	1
3	480	BACARDI USA INC	2024-01-12	8106	2023-12-20	2024-02-05	10100	1374
4	516	BANFI PRODUCTS CORP	2024-01-07	8170	2023-12-24	2024-02-12	1935	155



```
In [27]: engine
```

Out[27]: Engine(mysql+pymysql://root:***@localhost:3306/inventory)

```
In [41]: purchases = pd.read_sql("SELECT * FROM purchases LIMIT 5", engine)
```

```
In [40]: print(type(purchases))

<class 'pandas.DataFrame'>
```

```
In [42]: purchases = pd.read_sql("SELECT * FROM purchases LIMIT 5", engine)
```

```
In [43]: purchases.groupby(['Brand', 'PurchasePrice'])[['Quantity', 'Dollars']].sum()
```

Out[43]:

		Quantity	Dollars
--	--	----------	---------

Brand	PurchasePrice		
2034	21.32	5	106.60
5215	9.41	5	47.05
5255	9.35	10	93.50
8412	35.71	6	214.26

```
In [46]: vendor_invoice = pd.read_sql("SELECT * FROM vendor_invoice LIMIT 50", engine)
vendor_invoice
```

Out[46]:

	VendorNumber	VendorName	InvoiceDate	PONumber	PODate	PayDate	Quantity
0	105	ALTAMAR BRANDS LLC	2024-01-04	8124	2023-12-21	2024-02-16	6
1	4466	AMERICAN VINTAGE BEVERAGE	2024-01-07	8137	2023-12-22	2024-02-21	15
2	388	ATLANTIC IMPORTING COMPANY	2024-01-09	8169	2023-12-24	2024-02-16	5
3	480	BACARDI USA INC	2024-01-12	8106	2023-12-20	2024-02-05	10100
4	516	BANFI PRODUCTS CORP	2024-01-07	8170	2023-12-24	2024-02-12	1935
5	2396	BLACK PRINCE DISTILLERY INC	2024-01-08	8191	2023-12-25	2024-02-06	23
6	1128	BROWN-FORMAN CORP	2024-01-09	8150	2023-12-23	2024-02-19	4684
7	1189	BULLY BOY DISTILLERS	2024-01-09	8171	2023-12-24	2024-02-04	6
8	1273	CALEDONIA SPIRITS INC	2024-01-06	8172	2023-12-24	2024-02-15	5
9	11567	CAMPARI AMERICA	2024-01-06	8151	2023-12-23	2024-02-20	1321
10	90046	CANDIA VINEYARDS	2024-01-05	8107	2023-12-20	2024-02-10	24
11	1485	CASTLE BRANDS CORP.	2024-01-08	8152	2023-12-23	2024-02-19	320
12	2876	CENTEUR IMPORTS LLC	2024-01-08	8125	2023-12-21	2024-02-07	12
13	4380	CHARLES JACQUIN ET CIE INC	2024-01-06	8153	2023-12-23	2024-02-12	96
14	1392	CONSTELLATION BRANDS INC	2024-01-11	8108	2023-12-20	2024-02-10	8466
15	90047	CRUSH WINES	2024-01-07	8126	2023-12-21	2024-02-21	90
16	2242	DELICATO VINEYARDS INC	2024-01-06	8139	2023-12-22	2024-02-10	808

	VendorNumber	VendorName	InvoiceDate	PONumber	PODate	PayDate	Quantity
17	1590	DIAGEO CHATEAU ESTATE WINES	2024-01-12	8109	2023-12-20	2024-02-11	2246
18	3960	DIAGEO NORTH AMERICA INC	2024-01-10	8173	2023-12-24	2024-02-19	16602
19	2555	DISARONNO INTERNATIONAL LLC	2024-01-11	8192	2023-12-25	2024-02-17	385
20	2450	DJINN SPIRITS LLC	2024-01-11	8140	2023-12-22	2024-02-11	12
21	2496	DUGGANS DISTILLED PRODUCTS	2024-01-06	8174	2023-12-24	2024-02-10	24
22	3252	E & J GALLO WINERY	2024-01-09	8110	2023-12-20	2024-02-19	8086
23	2561	EDRINGTON AMERICAS	2024-01-08	8175	2023-12-24	2024-02-10	136
24	1655	FABRIZIA SPIRITS LLC	2024-01-04	8194	2023-12-25	2024-02-05	46
25	17031	FLAG HILL WINERY & VINEYARD	2024-01-06	8154	2023-12-23	2024-02-06	94
26	90011	FORTUNE WINE BROKERS LLC	2024-01-11	8141	2023-12-22	2024-02-14	23
27	9744	FREDERICK WILDMAN & SONS	2024-01-06	8176	2023-12-24	2024-02-12	1222
28	3924	HEAVEN HILL DISTILLERIES	2024-01-08	8155	2023-12-23	2024-02-15	818
29	3950	HOOD RIVER DISTILLERS, Inc.	2024-01-08	8195	2023-12-25	2024-02-07	6
30	90053	INCREDIBREW INC	2024-01-12	8178	2023-12-24	2024-02-12	48
31	17032	JEWELL TOWNE VINEYARDS	2024-01-10	8111	2023-12-20	2024-02-04	24
32	12546	JIM BEAM BRANDS COMPANY	2024-01-06	8142	2023-12-22	2024-02-19	11611
33	4550	KLIN SPIRITS LLC	2024-01-07	8196	2023-12-25	2024-02-17	7

	VendorNumber	VendorName	InvoiceDate	PONumber	PODate	PayDate	Quantity
34	4692	KOBRAND CORPORATION	2024-01-11	8112	2023-12-20	2024-02-06	427
35	90032	LABELLE VYDS AND WINERY	2024-01-06	8156	2023-12-23	2024-02-11	92
36	4848	LAIRD & CO	2024-01-07	8143	2023-12-22	2024-02-06	12
37	4950	LATITUDE BEVERAGE COMPANY	2024-01-07	8179	2023-12-24	2024-02-06	153
38	8352	LUXCO INC	2024-01-08	8113	2023-12-20	2024-02-07	819
39	9552	M S WALKER INC	2024-01-10	8157	2023-12-23	2024-02-09	8061
40	10000	MAJESTIC FINE WINES	2024-01-09	8128	2023-12-21	2024-02-17	875
41	5455	MARSALLE COMPANY	2024-01-08	8158	2023-12-23	2024-02-13	12
42	4425	MARTIGNETTI COMPANIES	2024-01-06	8160	2023-12-23	2024-02-06	24
43	4425	MARTIGNETTI COMPANIES	2024-01-05	8159	2023-12-23	2024-02-13	16075
44	5992	MCCORMICK DISTILLING CO	2024-01-10	8130	2023-12-21	2024-02-05	90
45	6213	MHW LTD	2024-01-09	8144	2023-12-22	2024-02-04	412
46	8112	MOET HENNESSY USA INC	2024-01-08	8180	2023-12-24	2024-02-06	1392
47	6215	MOONLIGHT MEADERY	2024-01-09	8161	2023-12-23	2024-02-10	35
48	6355	NICHE W & S	2024-01-05	8162	2023-12-23	2024-02-14	137
49	6359	OLE SMOKY DISTILLERY LLC	2024-01-10	8131	2023-12-21	2024-02-13	134

```
In [52]: #to check if the entry in the column is unique
vendor_invoice['PONumber'].nunique()
```

```
Out[52]: 50
```

```
In [64]: vendor_invoice.shape#if both nunique and the shape matches its unique
```

```
Out[64]: (50, 10)
```

```
In [66]: sales = pd.read_sql("SELECT * FROM sales LIMIT 30", engine)  
sales
```

Out[66]:

	InventoryId	Store	Brand	Description	Size	SalesQuantity	SalesDollars
0	1_HARDERSFIELD_1004	1	1004	Jim Beam w/2 Rocks Glasses	750mL	1	16.49
1	1_HARDERSFIELD_1004	1	1004	Jim Beam w/2 Rocks Glasses	750mL	2	32.98
2	1_HARDERSFIELD_1004	1	1004	Jim Beam w/2 Rocks Glasses	750mL	1	16.49
3	1_HARDERSFIELD_1004	1	1004	Jim Beam w/2 Rocks Glasses	750mL	1	14.49
4	1_HARDERSFIELD_1005	1	1005	Maker's Mark Combo Pack	375mL 2 Pk	2	69.98
5	1_HARDERSFIELD_1005	1	1005	Maker's Mark Combo Pack	375mL 2 Pk	1	34.99
6	1_HARDERSFIELD_1005	1	1005	Maker's Mark Combo Pack	375mL 2 Pk	1	34.99
7	1_HARDERSFIELD_1005	1	1005	Maker's Mark Combo Pack	375mL 2 Pk	1	34.99
8	1_HARDERSFIELD_10058	1	10058	F Coppola Dmd Ivry Cab Svgn	750mL	4	59.96
9	1_HARDERSFIELD_10058	1	10058	F Coppola Dmd Ivry Cab Svgn	750mL	1	14.99
10	1_HARDERSFIELD_10058	1	10058	F Coppola Dmd Ivry Cab Svgn	750mL	1	14.99
11	1_HARDERSFIELD_10058	1	10058	F Coppola Dmd Ivry Cab Svgn	750mL	1	14.99
12	1_HARDERSFIELD_10058	1	10058	F Coppola Dmd Ivry Cab Svgn	750mL	1	14.99

	InventoryId	Store	Brand	Description	Size	SalesQuantity	SalesDollars
13	1_HARDERSFIELD_10058	1	10058	F Coppola Dmd Ivry Cab Svgn	750mL	3	44.97
14	1_HARDERSFIELD_10058	1	10058	F Coppola Dmd Ivry Cab Svgn	750mL	2	29.98
15	1_HARDERSFIELD_10058	1	10058	F Coppola Dmd Ivry Cab Svgn	750mL	1	14.99
16	1_HARDERSFIELD_10058	1	10058	F Coppola Dmd Ivry Cab Svgn	750mL	1	14.99
17	1_HARDERSFIELD_1006	1	1006	Jim Beam Candy Cane 4/50mLs	50mL 4 Pk	1	3.99
18	1_HARDERSFIELD_10062	1	10062	Terroirs du Rhône	750mL	1	8.99
19	1_HARDERSFIELD_10062	1	10062	Terroirs du Rhône	750mL	1	8.99
20	1_HARDERSFIELD_10062	1	10062	Terroirs du Rhône	750mL	1	8.99
21	1_HARDERSFIELD_10062	1	10062	Terroirs du Rhône	750mL	1	8.99
22	1_HARDERSFIELD_10062	1	10062	Terroirs du Rhône	750mL	3	26.97
23	1_HARDERSFIELD_10062	1	10062	Terroirs du Rhône	750mL	1	8.99
24	1_HARDERSFIELD_10062	1	10062	Terroirs du Rhône	750mL	1	8.99
25	1_HARDERSFIELD_10062	1	10062	Terroirs du Rhône	750mL	1	8.99
26	1_HARDERSFIELD_10062	1	10062	Terroirs du Rhône	750mL	2	17.98
27	1_HARDERSFIELD_10062	1	10062	Terroirs du Rhône	750mL	1	8.99
28	1_HARDERSFIELD_10062	1	10062	Terroirs du Rhône	750mL	3	26.97
29	1_HARDERSFIELD_10062	1	10062	Terroirs du Rhône	750mL	1	8.99

```
In [67]: sales.groupby('Brand')[['SalesDollars', 'SalesPrice', 'SalesQuantity']].sum()
```

Out[67]:

	SalesDollars	SalesPrice	SalesQuantity
Brand			
1004	80.45	63.96	5
1005	174.95	139.96	5
1006	3.99	3.99	1
10058	224.85	134.91	15
10062	152.83	107.88	17

```
In [68]: vendor_invoice = pd.read_sql("SELECT * FROM vendor_invoice LIMIT 5", engine)
vendor_invoice
```

Out[68]:

	VendorNumber	VendorName	InvoiceDate	PONumber	PODate	PayDate	Quantity	
0	105	ALTAMAR BRANDS LLC	2024-01-04	8124	2023-12-21	2024-02-16	6	
1	4466	AMERICAN VINTAGE BEVERAGE	2024-01-07	8137	2023-12-22	2024-02-21	15	
2	388	ATLANTIC IMPORTING COMPANY	2024-01-09	8169	2023-12-24	2024-02-16	5	
3	480	BACARDI USA INC	2024-01-12	8106	2023-12-20	2024-02-05	10100	13
4	516	BANFI PRODUCTS CORP	2024-01-07	8170	2023-12-24	2024-02-12	1935	1

```
In [69]: vendor_invoice.columns
```

Out[69]: Index(['VendorNumber', 'VendorName', 'InvoiceDate', 'PONumber', 'PODate', 'PayDate', 'Quantity', 'Dollars', 'Freight', 'Approval'], dtype='str')

```
In [72]: freight_summary=pd.read_sql_query("""select VendorNumber,SUM(Freight) as FreightCos
From vendor_invoice
Group BY VendorNumber""",engine)
freight_summary
```

Out[72]:

	VendorNumber	FreightCost
0	105	62.39
1	4466	793.91
2	388	211.74
3	480	89286.27
4	516	8510.41
...	...	...
121	201359	0.09
122	4901	0.72
123	90059	74.84
124	5083	10.68
125	54	0.48

126 rows × 2 columns

In [77]:

```
query = """
SELECT
    p.VendorNumber,
    p.VendorName,
    p.Brand,
    p.PurchasePrice,
    pp.Volume,
    pp.Price AS ActualPrice,
    SUM(p.Quantity) AS TotalPurchaseQuantity,
    SUM(p.Dollars) AS TotalPurchaseDollars
FROM purchases p
JOIN purchase_prices pp
ON p.Brand = pp.Brand
where p.PurchasePrice>0
GROUP BY
    p.VendorNumber,
    p.VendorName,
    p.Brand,
    p.PurchasePrice,
    pp.Volume,
    pp.Price
ORDER BY TotalPurchaseDollars;
"""

result = pd.read_sql(query, engine)
result
```

Out[77]:

	VendorNumber	VendorName	Brand	PurchasePrice	Volume	ActualPrice	TotalPur
0	7245	PROXIMO SPIRITS INC.	3065	0.71	50	0.99	
1	3960	DIAGEO NORTH AMERICA INC	6127	1.47	200	1.99	
2	3924	HEAVEN HILL DISTILLERIES	9123	0.74	50	0.99	
3	8004	SAZERAC CO INC	5683	0.39	50	0.49	
4	9815	WINE GROUP INC	8527	1.32	750	4.99	
...	...	...	...	...	...	...	
10687	3960	DIAGEO NORTH AMERICA INC	3545	21.89	1750	29.99	
10688	3960	DIAGEO NORTH AMERICA INC	4261	16.17	1750	22.99	
10689	17035	PERNOD RICARD USA	8068	18.24	1750	24.99	
10690	4425	MARTIGNETTI COMPANIES	3405	23.19	1750	28.99	
10691	1128	BROWN-FORMAN CORP	1233	26.27	1750	36.99	

10692 rows × 8 columns

◀

▶

In [78]:

```
sales = pd.read_sql("SELECT * FROM sales LIMIT 5", engine)
sales
```

Out[78]:

	InventoryId	Store	Brand	Description	Size	SalesQuantity	SalesDollars	Sa
0	1_HARDERSFIELD_1004	1	1004	Jim Beam w/2 Rocks Glasses	750mL	1	16.49	
1	1_HARDERSFIELD_1004	1	1004	Jim Beam w/2 Rocks Glasses	750mL	2	32.98	
2	1_HARDERSFIELD_1004	1	1004	Jim Beam w/2 Rocks Glasses	750mL	1	16.49	
3	1_HARDERSFIELD_1004	1	1004	Jim Beam w/2 Rocks Glasses	750mL	1	14.49	
4	1_HARDERSFIELD_1005	1	1005	Maker's Mark Combo Pack	375mL 2 Pk	2	69.98	



In [80]: `sales.columns`

Out[80]: Index(['InventoryId', 'Store', 'Brand', 'Description', 'Size', 'SalesQuantity', 'SalesDollars', 'SalesPrice', 'SalesDate', 'Volume', 'Classification', 'ExciseTax', 'VendorNo', 'VendorName'], dtype='str')

In [91]: `pd.read_sql_query("""
SELECT
 VendorNo,
 Brand,
 SUM(SalesDollars) AS TotalSalesDollars,
 SUM(SalesPrice) AS TotalSalesPrice,
 SUM(SalesQuantity) AS TotalSalesQuantity,
 SUM(ExciseTax) AS TotalExciseTax
FROM sales
GROUP BY VendorNo, Brand
""", engine)`

Out[91]:

	VendorNo	Brand	TotalSalesDollars	TotalSalesPrice	TotalSalesQuantity	TotalExciseTax
0	12546	1004	720.52	426.22	48.0	37.8
1	12546	1005	28796.77	16865.18	823.0	322.7
2	2000	10058	160434.54	61141.54	11750.0	1313.0
3	12546	1006	1308.72	323.19	328.0	17.0
4	10754	10062	10365.69	5958.51	1131.0	125.6
...	...	...	...	...	...	...
9617	28776	3450	111.96	83.97	4.0	3.7
9618	10754	36071	1283.97	427.99	3.0	0.3
9619	10754	36072	245.97	81.99	3.0	0.3
9620	10754	36075	2096.97	698.99	3.0	0.3
9621	10754	36078	1223.97	407.99	3.0	0.3

9622 rows × 6 columns



In [90]:

```

vendor_sales_summary = pd.read_sql_query("""
WITH FreightSummary AS (
    SELECT
        VendorNumber,
        SUM(Freight) AS FreightCost
    FROM vendor_invoice
    GROUP BY VendorNumber
),
PurchaseSummary AS (
    SELECT
        p.VendorNumber,
        p.VendorName,
        p.Brand,
        p.Description,
        p.PurchasePrice,
        pp.Price AS ActualPrice,
        pp.Volume,
        SUM(p.Quantity) AS TotalPurchaseQuantity,
        SUM(p.Dollars) AS TotalPurchaseDollars
    FROM purchases p
    JOIN purchase_prices pp
    ON p.Brand = pp.Brand
    GROUP BY
        p.VendorNumber,
        p.VendorName,
        p.Brand,
        p.Description,
        p.PurchasePrice,
        pp.Price,

```

```

        pp.Volume
    ),

SalesSummary AS (
    SELECT
        VendorNo,
        Brand,
        SUM(SalesDollars) AS TotalSalesDollars,
        SUM(SalesPrice) AS TotalSalesPrice,
        SUM(SalesQuantity) AS TotalSalesQuantity,
        SUM(ExciseTax) AS TotalExciseTax
    FROM sales
    GROUP BY VendorNo, Brand
)

SELECT
    ps.VendorNumber,
    ps.VendorName,
    ps.Brand,
    ps.Description,
    ps.PurchasePrice,
    ps.ActualPrice,
    ps.Volume,
    ps.TotalPurchaseQuantity,
    ps.TotalPurchaseDollars,
    ss.TotalSalesQuantity,
    ss.TotalSalesDollars,
    ss.TotalSalesPrice,
    ss.TotalExciseTax,
    fs.FreightCost
FROM PurchaseSummary ps
LEFT JOIN SalesSummary ss
    ON ps.VendorNumber = ss.VendorNo
    AND ps.Brand = ss.Brand
LEFT JOIN FreightSummary fs
    ON ps.VendorNumber = fs.VendorNumber
ORDER BY ps.TotalPurchaseDollars DESC;
""", engine)

vendor_sales_summary.head()

```

Out[90]:

	VendorNumber	VendorName	Brand	Description	PurchasePrice	ActualPrice	Volume
0	1128	BROWN-FORMAN CORP	1233	Jack Daniels No 7 Black	26.27	36.99	1750
1	4425	MARTIGNETTI COMPANIES	3405	Tito's Handmade Vodka	23.19	28.99	1750
2	17035	PERNOD RICARD USA	8068	Absolut 80 Proof	18.24	24.99	1750
3	3960	DIAGEO NORTH AMERICA INC	4261	Capt Morgan Spiced Rum	16.17	22.99	1750
4	3960	DIAGEO NORTH AMERICA INC	3545	Ketel One Vodka	21.89	29.99	1750

In [92]:

vendor_sales_summary.isnull().sum()

Out[92]:

VendorNumber0
VendorName0
Brand0
Description0
PurchasePrice0
ActualPrice0
Volume0
TotalPurchaseQuantity0
TotalPurchaseDollars0
TotalSalesQuantity1789
TotalSalesDollars1789
TotalSalesPrice1789
TotalExciseTax1789
FreightCost0
dtype: int64

In [99]:

vendor_sales_summary.dtypes

```
Out[99]: VendorNumber      int64
VendorName      str
Brand      int64
Description      str
PurchasePrice      float64
ActualPrice      float64
Volume      float64
TotalPurchaseQuantity      float64
TotalPurchaseDollars      float64
TotalSalesQuantity      float64
TotalSalesDollars      float64
TotalSalesPrice      float64
TotalExciseTax      float64
FreightCost      float64
dtype: object
```

```
In [100... vendor_sales_summary['Volume'] = vendor_sales_summary['Volume'].astype('float64')
```

```
In [97]: vendor_sales_summary.fillna(0, inplace=True)
```

Out[97]:

	VendorNumber	VendorName	Brand	Description	PurchasePrice	ActualPrice	Vo
0	1128	BROWN-FORMAN CORP	1233	Jack Daniels No 7 Black	26.27	36.99	1
1	4425	MARTIGNETTI COMPANIES	3405	Tito's Handmade Vodka	23.19	28.99	1
2	17035	PERNOD RICARD USA	8068	Absolut 80 Proof	18.24	24.99	1
3	3960	DIAGEO NORTH AMERICA INC	4261	Capt Morgan Spiced Rum	16.17	22.99	1
4	3960	DIAGEO NORTH AMERICA INC	3545	Ketel One Vodka	21.89	29.99	1
...	...	...	...	...	...	...	...
10688	8004	SAZERAC CO INC	5683	Dr McGillicuddy's Apple Pie	0.39	0.49	
10689	3924	HEAVEN HILL DISTILLERIES	9123	Deep Eddy Vodka	0.74	0.99	
10690	3960	DIAGEO NORTH AMERICA INC	6127	The Club Strawbry Margarita	1.47	1.99	
10691	7245	PROXIMO SPIRITS INC.	3065	Three Olives Grape Vodka	0.71	0.99	
10692	2561	EDRINGTON AMERICAS	2166	The Macallan Double Cask 12	0.00	0.00	

10693 rows × 14 columns



```
In [98]: vendor_sales_summary['VendorName'] = vendor_sales_summary['VendorName'].str.strip()
```

```
In [101... vendor_sales_summary['GrossProfit'] = vendor_sales_summary['TotalSalesDollars'] - v
```

```
In [104... vendor_sales_summary['GrossProfit'].min()
```

```
Out[104... np.float64(-1751843.6800000893)
```

```
In [105... vendor_sales_summary['ProfitMargin'] = (vendor_sales_summary['GrossProfit'] / v
```

```
In [106... vendor_sales_summary['StockTurnover'] = vendor_sales_summary['TotalSalesQuantity']
```

```
In [107... vendor_sales_summary['SalestoPurchaseRatio'] = vendor_sales_summary['TotalSalesDoll
```

```
In [116... import numpy as np

vendor_sales_summary.replace([np.inf, -np.inf], 0, inplace=True)

vendor_sales_summary.to_sql(
    'vendor_sales_summary',
    con=engine,
    if_exists='replace',
    index=False
)
```

```
Out[116... 10693
```

```
In [117... pd.read_sql_query("SELECT * FROM vendor_sales_summary", engine)
```

Out[117...

	VendorNumber	VendorName	Brand	Description	PurchasePrice	ActualPrice	Vo
0	1128	BROWN-FORMAN CORP	1233	Jack Daniels No 7 Black	26.27	36.99	1
1	4425	MARTIGNETTI COMPANIES	3405	Tito's Handmade Vodka	23.19	28.99	1
2	17035	PERNOD RICARD USA	8068	Absolut 80 Proof	18.24	24.99	1
3	3960	DIAGEO NORTH AMERICA INC	4261	Capt Morgan Spiced Rum	16.17	22.99	1
4	3960	DIAGEO NORTH AMERICA INC	3545	Ketel One Vodka	21.89	29.99	1
...	...	...	...	...	...	...	...
10688	8004	SAZERAC CO INC	5683	Dr McGillicuddy's Apple Pie	0.39	0.49	
10689	3924	HEAVEN HILL DISTILLERIES	9123	Deep Eddy Vodka	0.74	0.99	
10690	3960	DIAGEO NORTH AMERICA INC	6127	The Club Strawbry Margarita	1.47	1.99	
10691	7245	PROXIMO SPIRITS INC.	3065	Three Olives Grape Vodka	0.71	0.99	
10692	2561	EDRINGTON AMERICAS	2166	The Macallan Double Cask 12	0.00	0.00	

10693 rows × 18 columns



In [118...

```
import numpy as np
# Check the data
vendor_sales_summary.info()
vendor_sales_summary.head()
# Remove leading and trailing spaces
vendor_sales_summary['VendorName'] = vendor_sales_summary['VendorName'].str.strip()
# Convert Volume to float
vendor_sales_summary['Volume'] = vendor_sales_summary['Volume'].astype('float64')
# Replace infinite values with NaN
vendor_sales_summary.replace([np.inf, -np.inf], np.nan, inplace=True)
# Fill missing values with 0
vendor_sales_summary.fillna(0, inplace=True)
# Remove duplicate rows (if any)
```

```

vendor_sales_summary.drop_duplicates(inplace=True)
# Verify the cleaned data
vendor_sales_summary.info()
vendor_sales_summary.head()

```

```

<class 'pandas.DataFrame'>
RangeIndex: 10693 entries, 0 to 10692
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   VendorNumber                          10693 non-null  int64
1   VendorName                            10693 non-null  str
2   Brand                                10693 non-null  int64
3   Description                           10693 non-null  str
4   PurchasePrice                         10693 non-null  float64
5   ActualPrice                           10693 non-null  float64
6   Volume                                10693 non-null  float64
7   TotalPurchaseQuantity                 10693 non-null  float64
8   TotalPurchaseDollars                  10693 non-null  float64
9   TotalSalesQuantity                   10693 non-null  float64
10  TotalSalesDollars                     10693 non-null  float64
11  TotalSalesPrice                       10693 non-null  float64
12  TotalExciseTax                        10693 non-null  float64
13  FreightCost                           10693 non-null  float64
14  GrossProfit                           10693 non-null  float64
15  ProfitMargin                          10692 non-null  float64
16  StockTurnover                         10693 non-null  float64
17  SalestoPurchaseRatio                  10692 non-null  float64
dtypes: float64(14), int64(2), str(2)
memory usage: 1.5 MB
<class 'pandas.DataFrame'>
RangeIndex: 10693 entries, 0 to 10692
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   VendorNumber                          10693 non-null  int64
1   VendorName                            10693 non-null  str
2   Brand                                10693 non-null  int64
3   Description                           10693 non-null  str
4   PurchasePrice                         10693 non-null  float64
5   ActualPrice                           10693 non-null  float64
6   Volume                                10693 non-null  float64
7   TotalPurchaseQuantity                 10693 non-null  float64
8   TotalPurchaseDollars                  10693 non-null  float64
9   TotalSalesQuantity                   10693 non-null  float64
10  TotalSalesDollars                     10693 non-null  float64
11  TotalSalesPrice                       10693 non-null  float64
12  TotalExciseTax                        10693 non-null  float64
13  FreightCost                           10693 non-null  float64
14  GrossProfit                           10693 non-null  float64
15  ProfitMargin                          10693 non-null  float64
16  StockTurnover                         10693 non-null  float64
17  SalestoPurchaseRatio                  10693 non-null  float64
dtypes: float64(14), int64(2), str(2)
memory usage: 1.5 MB

```

Out[118...

	VendorNumber	VendorName	Brand	Description	PurchasePrice	ActualPrice	Volume
0	1128	BROWN-FORMAN CORP	1233	Jack Daniels No 7 Black	26.27	36.99	1750.0
1	4425	MARTIGNETTI COMPANIES	3405	Tito's Handmade Vodka	23.19	28.99	1750.0
2	17035	PERNOD RICARD USA	8068	Absolut 80 Proof	18.24	24.99	1750.0
3	3960	DIAGEO NORTH AMERICA INC	4261	Capt Morgan Spiced Rum	16.17	22.99	1750.0
4	3960	DIAGEO NORTH AMERICA INC	3545	Ketel One Vodka	21.89	29.99	1750.0

In [119...

```
import logging
logging.basicConfig(level=logging.INFO)
logging.info("Cleaning Data.....")
vendor_sales_summary['VendorName'] = vendor_sales_summary['VendorName'].str.strip()
vendor_sales_summary['Volume'] = vendor_sales_summary['Volume'].astype('float64')
vendor_sales_summary.replace([np.inf, -np.inf], np.nan, inplace=True)
vendor_sales_summary.fillna(0, inplace=True)
vendor_sales_summary.drop_duplicates(inplace=True)
logging.info("Cleaning Completed.")
logging.info(vendor_sales_summary.head())
```

INFO:root:Cleaning Data.....

INFO:root:Cleaning Completed.

INFO:root:	VendorNumber		VendorName	Brand	Description
\					
0	1128	BROWN-FORMAN CORP	1233	Jack Daniels No 7 Black	
1	4425	MARTIGNETTI COMPANIES	3405	Tito's Handmade Vodka	
2	17035	PERNOD RICARD USA	8068	Absolut 80 Proof	
3	3960	DIAGEO NORTH AMERICA INC	4261	Capt Morgan Spiced Rum	
4	3960	DIAGEO NORTH AMERICA INC	3545	Ketel One Vodka	

	PurchasePrice	ActualPrice	Volume	TotalPurchaseQuantity	\
0	26.27	36.99	1750.0	145080.0	
1	23.19	28.99	1750.0	164038.0	
2	18.24	24.99	1750.0	187407.0	
3	16.17	22.99	1750.0	201682.0	
4	21.89	29.99	1750.0	138109.0	

	TotalPurchaseDollars	TotalSalesQuantity	TotalSalesDollars	\
0	3811251.60	65589.0	2362469.11	
1	3804041.22	67842.0	2052197.54	
2	3418303.68	87154.0	2142794.96	
3	3261197.94	99968.0	2249771.32	
4	3023206.01	64482.0	2007557.18	

	TotalSalesPrice	TotalExciseTax	FreightCost	GrossProfit	ProfitMargin	\
0	355827.73	120511.45	68601.68	-1448782.49	-61.324928	
1	295191.71	124652.04	144929.24	-1751843.68	-85.364281	
2	246754.00	160137.15	123780.22	-1275508.72	-59.525468	
3	225580.43	183683.43	257032.07	-1011426.62	-44.956863	
4	285284.03	118478.23	257032.07	-1015648.83	-50.591278	

	StockTurnover	SalestoPurchaseRatio
0	0.452089	0.619867
1	0.413575	0.539478
2	0.465052	0.626859
3	0.495671	0.689860
4	0.466892	0.664049

In []: